

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: 26067

Problem Statement Title: Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.

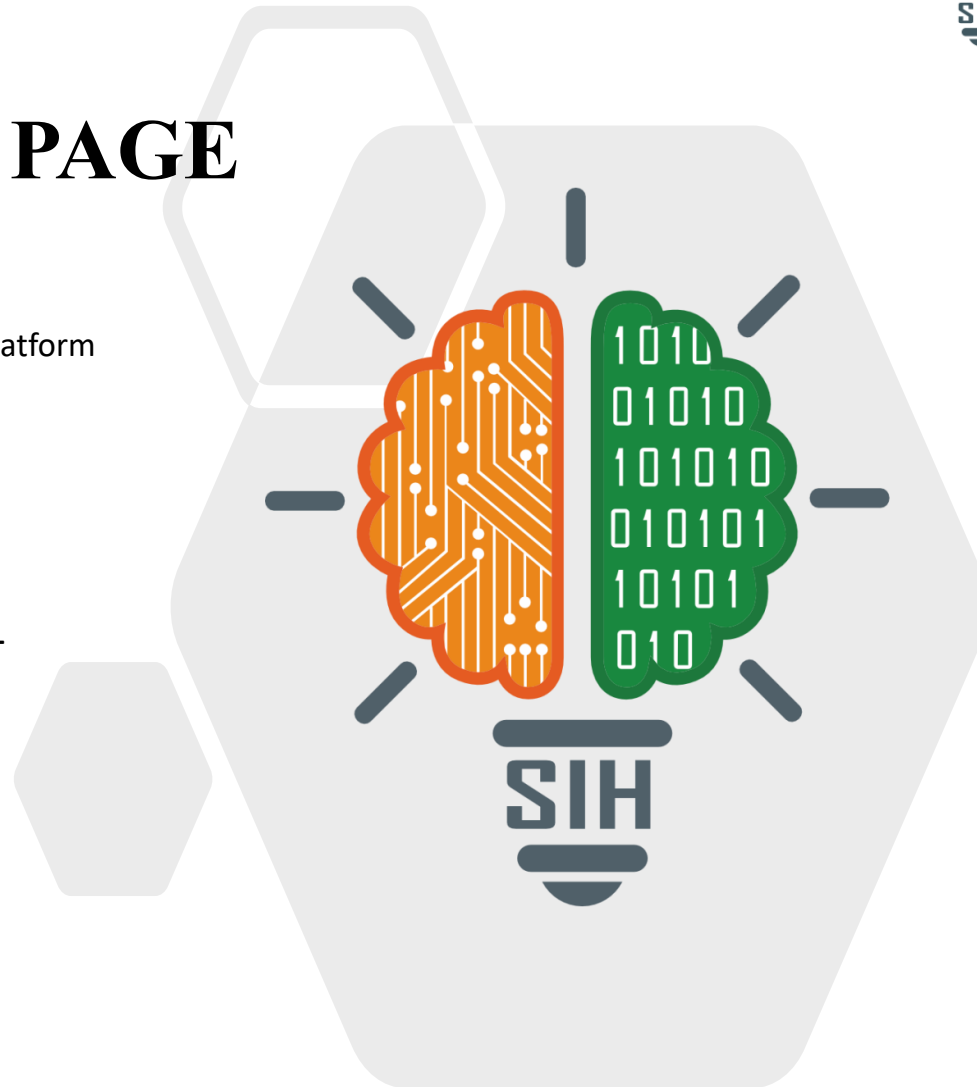
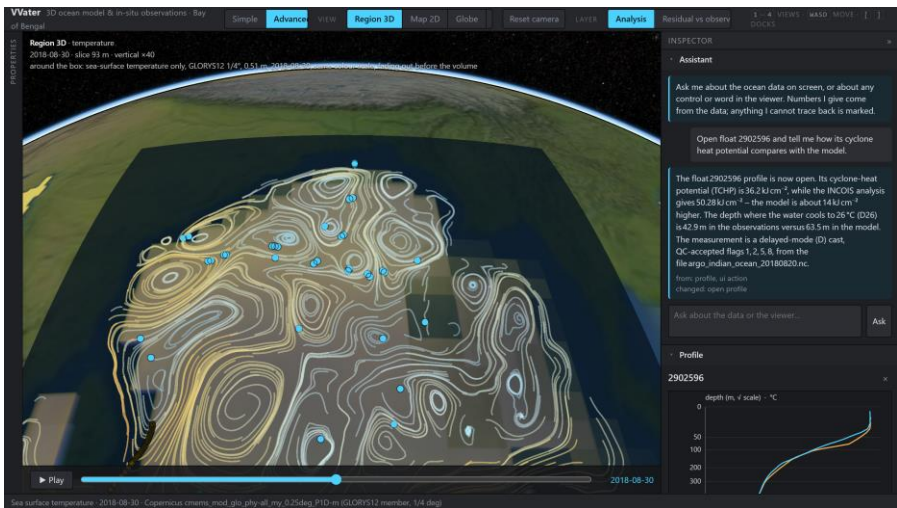
Theme: Disaster Management

PS Category: Software

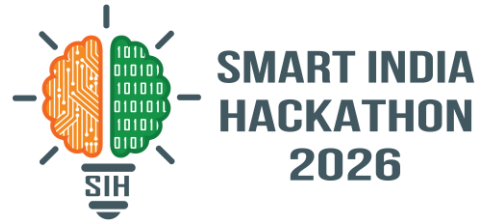
Team ID:

Team Name: VWater

The ocean model and every float and glider in one 3D scene, in a browser — and where they disagree.



VWater: Ocean in 3D



VWater

INCOIS knows the ocean in 3D; its forecasters still read it in 2D. **VWater puts the model and every instrument in one 3D scene, in a browser, and shows where they disagree.**

THE PROBLEM, IN PLAIN WORDS

The ocean has depth

A cyclone feeds on a warm layer tens of metres thick. A surface map cannot show how thick, and thickness is the fuel.

Two versions of truth

The **model** estimates the whole ocean on a grid; **floats and gliders** actually measure, at a few points. Where they disagree matters most.

Two separate tools

Each is opened in its own desktop program, mostly as flat maps, so comparing them is manual — the gap the brief names.

ONE QUESTION, TODAY AND WITH VVATER

Today	Open the model in one desktop tool, pick a depth	Open the float files in another, plot one profile	Line them up by eye; repeat for every depth
VWater	Open one link. No install ; deployable on INCOIS servers	Whole water column + 35 float profiles , 112 glider casts	Click a float: profile vs model, gap in °C and cyclone fuel

WHAT MAKES IT DIFFERENT

Shows where the model is wrong

Measured minus model as its own 3D layer. Only **4.86%** was ever measured; the rest stays visibly empty.

Shows how sure the model is

The analysis ships an error field; water the model is guessing about is drawn faint, not solid.

Speaks the forecaster's unit

Every comparison also in **cyclone heat potential** (kJ/cm²); the 20 °C isotherm is one click.

Never hides a bad reading

Each point carries its QC flag, data mode and source file; rejected levels are drawn red, not deleted.

Argo float

a robot that dives to 2,000 m every ten days

Glider

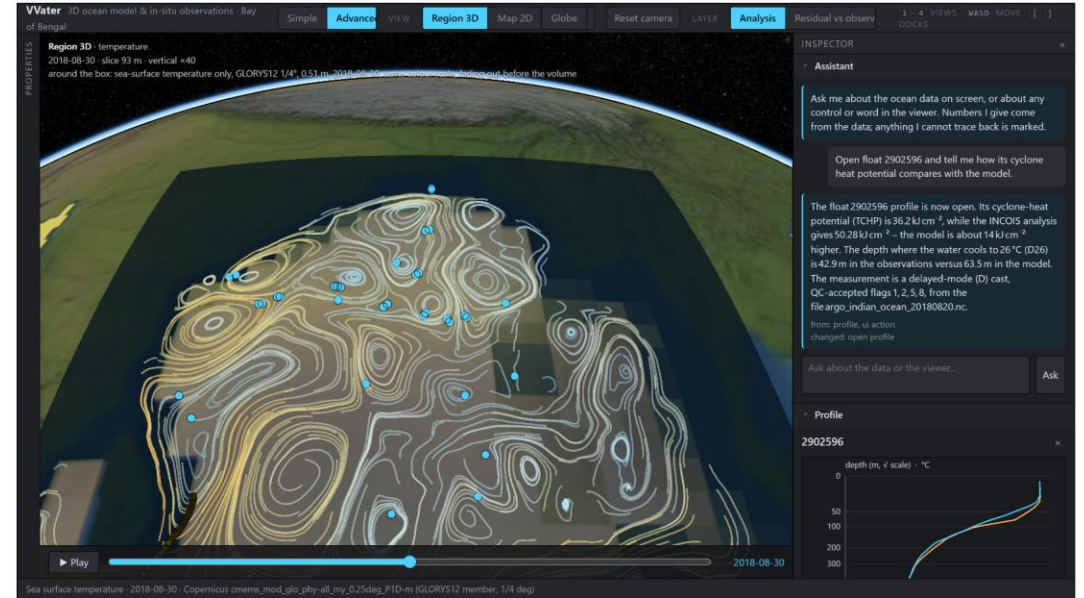
an underwater drone that saw-tooths along a track

Analysis

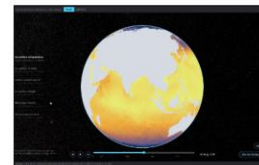
the model's best 3D estimate, published by INCOIS

Residual

measured minus modelled: how wrong, and where



The working prototype, not a mock-up: the volume cut at 93 m with currents and every float; right, the built-in assistant opened a float for us and compared it with the model from the data.



Simple

the whole ocean, one layer



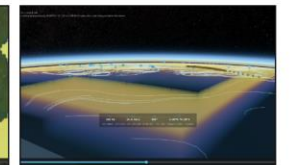
Region 3D

water column, 40× tall



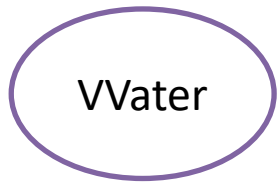
Map 2D

section at any depth



Fly

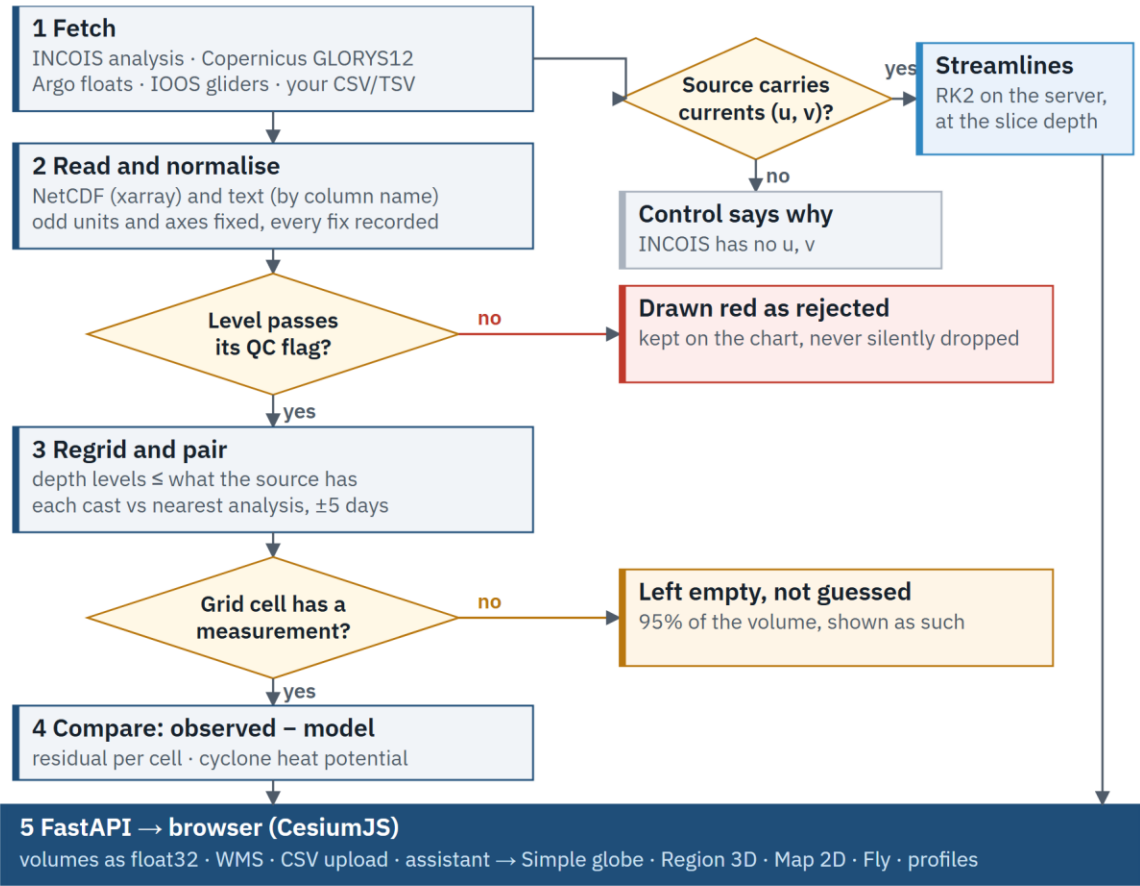
over the Bay, fixed height



TECHNICAL APPROACH



FROM RAW FILES TO THE SCENE – AND WHAT HAPPENS TO BAD OR MISSING DATA



Diamonds are decisions the code makes. The red and amber branches are where most tools quietly drop a bad reading or paint over a gap; here both stay on screen, labelled.

STACK – EACH CHOICE AGAINST WHAT WE DID NOT PICK

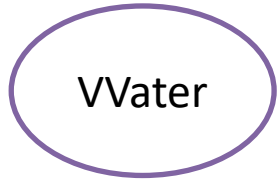
CHOSEN	INSTEAD OF	WHY
CesiumJS	Three.js, deck.gl	A real globe with correct geography, and a built-in voxel raymarcher; we wrote only the colour transfer function. Three.js needs both hand-built; deck.gl has no volume layer.
TypeScript + Vite	React, Angular	No UI framework, on purpose: one Cesium canvas and a control panel. React or Angular would add a second render loop beside Cesium's; typed ES modules do the job.
FastAPI	Flask, Django	The science stack (xarray, NumPy, TEOS-10) is Python; typed parameters and API docs come free; nothing here needs Django's database layer.
xarray	PyNIO	The brief names both; PyNIO is archived and unmaintained, xarray is live and reads CF metadata.
Raw float32	JSON, OPeNDAP	Bytes go straight into a GPU texture with no parsing: one full INCOIS timestep is 0.55 MB .
OGC WMS 1.3.0	only our own tiles	The same layers open in QGIS or any national portal, as the brief's open-standards clause asks.
Groq LLM + tools	a plain chatbot	The model can only answer through our own API functions; every number in a reply is checked against the data it came from, and untraceable ones are flagged on screen.

HOW A VOLUME IS DRAWN

Opacity per metre of water, not per sample
so a coarse and a fine grid of the same sea look the same

Depth levels spaced by $\sqrt{\text{depth}}$, never more than the source has
24 levels, 5–2000 m; inventing levels would invent a thermocline

New sensor = one parser function and one line of config
the same glider as CSV and as NetCDF gives 222 of 222 casts identical



FEASIBILITY AND VIABILITY



35

Argo profiles in the demo window, 33 scientist-checked (delayed mode)

112

glider casts, 13,053 levels, from an independent instrument

0.55 MB

one full 3D timestep, temperature and its error, as sent to the browser

0.45 s

to pair 127 casts with the model and build the residual volume

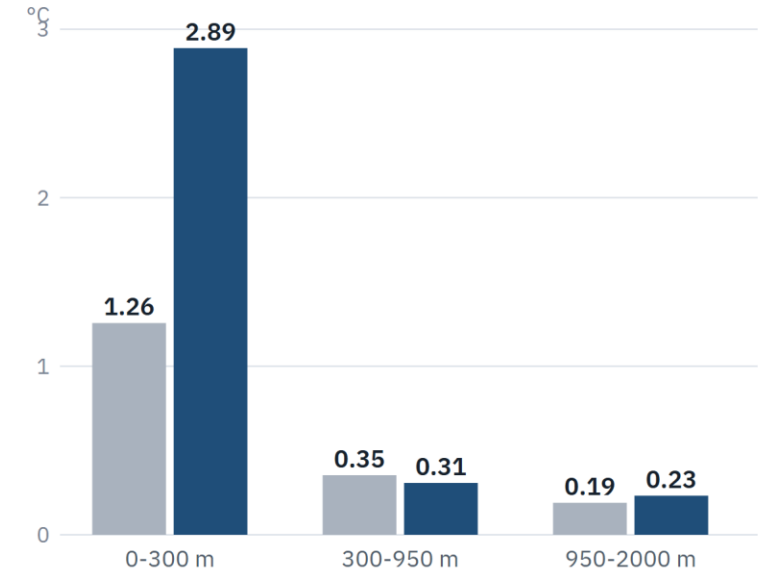
397

current streamlines integrated in 397 ms on the server

222/222

casts identical read from CSV and from NetCDF — 24,611 levels

MEASURED FINDING: MODEL ERROR BY DEPTH



■ Argo (the model already used them) ■ Glider (independent)

Below 300 m model and instruments agree (0.35 vs 0.31 °C). In the top 300 m — the warm layer a cyclone feeds on — the independent glider finds **2.89 °C**. That is the gap this tool makes visible.

RUNS ON WHAT INCOIS HAS

One Python process + static files
no database server, no container required

Any modern browser
no plug-in, no install, no account

No third-party service at demo time
offline basemap, no map-tile token, data cached after first fetch

INCOIS data needs no credentials
public ERDDAP; Copernicus is optional, for currents

Graphics tuned to the machine
four quality tiers, picked by measuring real frames at start-up

Every claim reproducible
one command regenerates every number on these slides

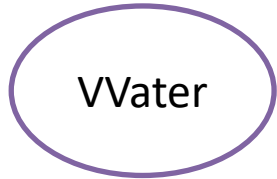
MODEL VS TWO INSTRUMENTS

AGAINST	CASTS	BIAS	RMSE
Argo	29	−0.20 °C	0.71 °C
Glider	98	−0.69 °C	1.58 °C

Argo is a consistency check (the model assimilates it); the glider is the independent test.

RISKS WE HIT, AND WHAT WE DID

RISK	HANDLED BY
Brief's data links are dead	Both are ftp://; port 21 is blocked. Replaced with tested HTTPS mirrors.
Impossible values in the model	6 of 68,022 cells fail the Argo range test; masked and counted, not drawn.
Files not quite CF-standard	Units like degs read defensively; each assumption shown with the data.
Bad float levels	510 of 11,525 levels rejected by QC; drawn red, never dropped.
Volume API is experimental	Cesium version pinned exactly; re-tested on each upgrade.
INCOIS TLS chain incomplete	OS trust store, not disabled verification.
Grid too large for a browser	Display-only block mean, factor recorded; depth never resampled up.
Copernicus needs an account	INCOIS is the default and needs none; only currents use Copernicus.
Unknown GPUs at INCOIS	Quality tiers, chosen by measuring real frames on that machine.
A cast with no model value	Profile still drawn, with the reason printed; nothing filled in.



IMPACT AND BENEFITS



FOR THE CYCLONE FORECASTER

+16.0

kJ/cm^2 — model minus measured **cyclone heat potential** along the independent glider track (95 casts, one deployment; measured mean 9.5). Against Argo, which the model already absorbs: -3.5 .

Fuel, in the unit already used

heat above 26°C , per float and per glider cast

20 °C isotherm in one click

the standard proxy for how deep the warm layer goes

Model and truth side by side

no second program, no copying numbers between tools

Confidence visible

uncertain water drawn faint; unmeasured water drawn empty

FOR INCOIS OPERATIONS

Search and rescue, fisheries, climate

currents and temperature at any depth, in the same view

New instruments without re-engineering

drop a CSV of casts on the globe; it is paired with the model at once

Open standards

WMS layers open in QGIS and national portals

FOR STUDENTS, PUBLIC AND POLICY

4.86%

of the Bay's water column had a measurement in this window. Shown honestly, that is itself the case for more floats — a sentence a policymaker can repeat.

True scale in one click

the Bay really is a film of water: 2 km deep, 2,400 km wide. Then 40 $\times$, to read it

From planet to profile

a simple globe, then the Bay in 3D, then fly over it, then one float's dive

Runs in a school's browser

exhibitions, e-learning and outreach, as the brief asks

Ask it in plain words

a built-in assistant answers from the data and moves the view for you

WHERE IT GOES NEXT

More regions and years

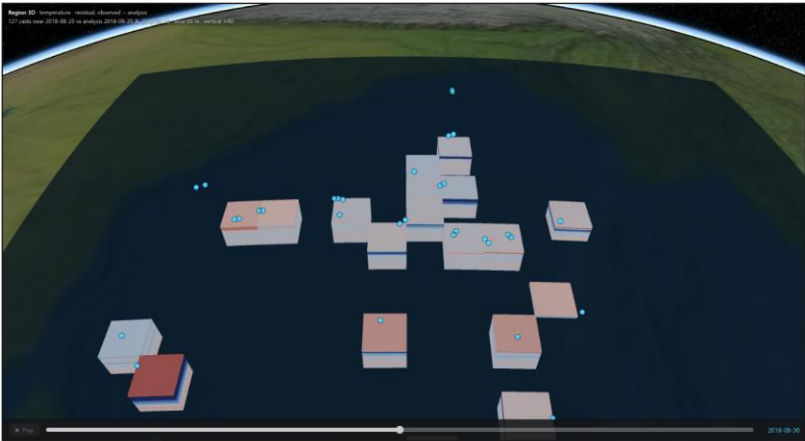
the region and window are configuration, not code

More sensors

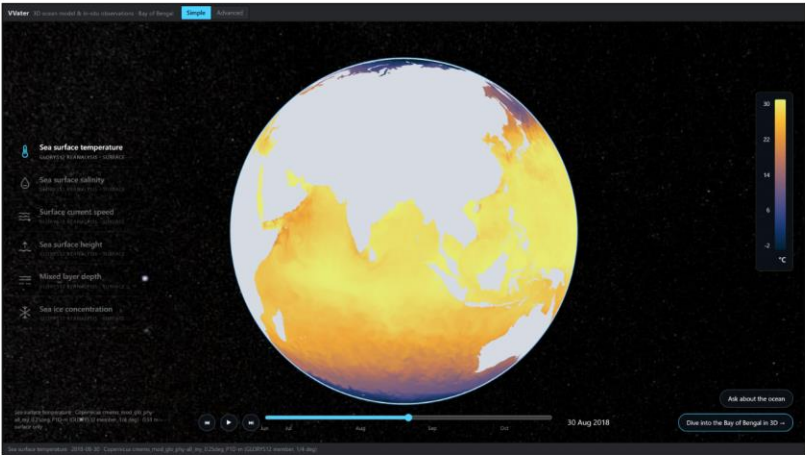
CTD, moorings, HF radar, ADCP: each one parser

Machine-learning products

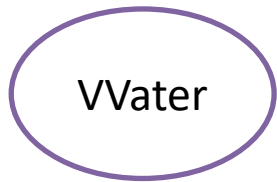
any gridded output renders as another volume



Where the model is wrong. Each block is a place someone measured, coloured by measured minus model (red warmer, blue colder). Everything else is empty on purpose: 95% of the volume.



The Simple view, for everyone else. The world's ocean one layer at a time — temperature, salinity, currents, sea level, mixed layer, ice — on a timeline.



RESEARCH AND REFERENCES



Source code, docs and every measured number: <https://github.com/Kukyos/VWater>

public repository · all links below opened 2026-09-23

DATA WE FETCHED — EACH ONE PROBED, NOT ASSUMED

- 1 INCOIS ERDDAP — Argo 10-day variational analysis ([incois_argo_10d_VAM](https://incois-argo-10d-VAM)), temperature and salinity with per-cell error. The primary volume; no credentials.
erddap.incois.gov.in/erddap/griddap/incois_argo_10d_VAM.html
- 2 Copernicus Marine — GLORYS12 global reanalysis, GLOBAL_MULTIYEAR_PHY_001_030. Currents, and the global surface layer.
data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030
- 3 Argo Global Data Assembly Centre, HTTPS mirror. Float profiles with QC flags and data mode.
data-argo.ifremer.fr
- 4 U.S. IOOS Glider DAC — Rutgers *ru29* deployment, NetCDF and CSV. The independent check.
gliders.ioos.us/erddap
- 5 INCOIS Live Access Server. Named in the brief; reachable, not yet ingested.
las.incois.gov.in/las
- 6 The brief's two ftp.ifremer.fr links. Probed: do not connect (port 21). Replaced by 3.
ftp://ftp.ifremer.fr/ifremer/argo · [.../glider/v2](https://ftp://ftp.ifremer.fr/ifremer/v2)
- 7 Argo Program overview, Scripps Institution of Oceanography. What a float is and does.
argo.ucsd.edu
- 8 Smart India Hackathon 2026, problem statement 26067 (MoES / INCOIS Ocean Valley). Quoted verbatim in the repository.
github.com/Kukyos/VWater/blob/main/docs/01-problem-statement.md

How these links were checked

Every URL was opened on 2026-09-23 with a browser user agent and returned HTTP 200, except the two FTP links (dead, listed as such) and the AMS journal page, which refuses scripted requests; its DOI resolves.

STANDARDS AND METHODS THE CODE FOLLOWS

- 9 Wong et al., Argo Quality Control Manual for CTD and Trajectory Data. Range test and QC flags.
doi.org/10.13155/33951
- 10 U.S. IOOS QARTOD manuals. Glider QC flags.
ioos.noaa.gov/project/qartod
- 11 NetCDF Climate and Forecast (CF) Conventions. Read defensively; assumptions recorded.
cfconventions.org
- 12 OGC Web Map Service 1.3.0. The WMS endpoint.
ogc.org/standards/wms
- 13 IOC, SCOR, IAPSO (2010), TEOS-10 seawater equation. Pressure to depth, heat capacity.
teos-10.org
- 14 Leipper & Volgenau (1972), Hurricane heat potential of the Gulf of Mexico, J. Phys. Oceanogr. 2. Heat above 26 °C.
[doi.org/10.1175/1520-0485\(1972\)002<0218:HHPOTG>2.0.CO;2](https://doi.org/10.1175/1520-0485(1972)002<0218:HHPOTG>2.0.CO;2)
- 15 Thyng et al. (2016), True colors of oceanography, Oceanography 29(3). cmocan palettes.
doi.org/10.5670/oceanog.2016.66
- 16 CesiumJS VoxelPrimitive reference. The volume renderer (experimental, pinned).
cesium.com/learn/cesiumjs/ref-doc/VoxelPrimitive.html
- 17 xarray and FastAPI documentation. Ingest and API.
docs.xarray.dev · fastapi.tiangolo.com
- 18 Unidata NetCDF. The file format the brief names.
unidata.ucar.edu/software/netcdf